TEXORA · COURSE NOTES · LECTURE SHEET

The Industrial Revolution

The change from hand production to machine production in factories — its stages, and what it did to the economy, to society, to the family and to work itself.

HUM-2209 · SOCIOLOGY AND HUMAN ECOLOGY · TOPIC 3 · LEVEL 2 TERM 2

1. Definition and beginnings

The **Industrial Revolution** is the transformation, beginning in Britain in the later eighteenth century, by which goods came to be made by machines driven by inanimate power in factories, instead of by hand tools in homes and small workshops. “Revolution” is the right word despite the change taking decades: it altered not only how things were made but where people lived, how families earned, what classes existed and what daily time felt like.

It began in Britain for a combination of reasons historians still weigh: rich coal and iron deposits, capital accumulated from trade, colonies supplying raw cotton and captive markets, a growing population supplying labour, patent protection and a practical inventive tradition, and canals and ports for cheap transport. Crucially for a textile student, **the first industry to be mechanised was cotton textiles** — the industry this degree is about. A chain of inventions in spinning and weaving — the flying shuttle, the spinning jenny, the water frame, the mule and the power loom — made hand production uncompetitive within a generation, and steam power (perfected by James Watt) freed the factory from the riverbank.

2. Stages of the Industrial Revolution

STAGE	ROUGH PERIOD	ENERGY AND KEY INVENTIONS	PRODUCTION SYSTEM
First	c. 1760–1840	Steam engine; mechanised spinning and weaving; iron	The factory replaces the cottage; first in textiles, then mining and transport (railways, steamships)
Second	c. 1870–1914	Electricity, oil, cheap steel; telegraph and telephone; internal combustion engine	Mass production and the assembly line; giant firms, railways and cities; science applied systematically to industry

Third	c. 1970–2000	Computers, electronics, programmable automation	Automated lines, computer-aided design and manufacture; information becomes a raw material
Fourth (“Industry 4.0”)	21st century	Internet of things, robotics, artificial intelligence, big data	The “smart factory”: machines sense, communicate and adjust; the boundary between making and computing blurs

The stages are a convenient map, not four separate events: each built on the previous one's skills, capital and institutions, and in much of the world — Bangladesh included — several stages coexist, with hand processes working beside automated lines in the same industrial district.

3. Effects on the economy

- **The factory system** — production concentrated in one building under one employer's discipline, replacing the domestic system in which merchants put out work to homes.
- **Mass production and cheap goods** — output multiplied and prices fell; cloth, once a luxury, became ordinary.
- **Capitalism and new organisation** — large amounts of capital had to be gathered before production began, giving rise to joint-stock companies, banks and stock exchanges in their modern form.
- **Division of labour** — work was broken into narrow repeated operations, raising productivity enormously (Adam Smith's pin factory is the classic description).
- **World markets and trade** — industrial nations hunted raw materials and markets worldwide; colonies were reorganised to supply both, a pattern whose marks on economies like Bengal's — once famous for its muslin, reduced to exporting raw material — are still studied.
- **New economic problems** — trade cycles of boom and slump, unemployment from displaced hand workers, and industrial pollution arrived with the factory and have never left.

4. Effects on society

- **Urbanization.** Workers crowded to factory towns; cities grew faster than housing, water or sanitation — producing the slum, the epidemic and eventually the modern sciences of public health and town planning.
- **New classes.** Society reorganised around two great new classes: the industrial *bourgeoisie* owning capital, and the *proletariat* owning only its labour. An intermediate middle class of managers, clerks and professionals grew between them (Topic 4 takes up this structure).

- **Working conditions.** Early factories imposed very long hours, child labour, dangerous machinery and harsh discipline. Reaction produced factory laws, limits on child labour, and the trade union movement — the organised voice of workers.
- **Population growth.** Cheaper food, later better medicine and sanitation, let populations rise steeply (Topic 6).
- **New ideas.** Laissez-faire liberalism, socialism and Marxism were all, in part, arguments about what the Industrial Revolution had done and what should be done about it.

5. Effects on the family and on work

AREA	BEFORE (AGRARIAN / DOMESTIC SYSTEM)	AFTER (INDUSTRIAL SYSTEM)
Place of work	The home and its fields; family works together	The factory and office, away from home; work and family life separated in place and time
Family type	Large joint/extended family, an economic unit	Smaller nuclear family fits mobile industrial life; extended kin weaken
Women and children	Worked within the household economy	Drawn into factories (early exploitation), later into offices and professions — the beginning of women's economic independence and of debates on childcare
Skill and the worker	The craftsman owns tools, controls pace, sees the whole product	The machine sets the pace; narrow tasks deskill many jobs; Marx called the worker's resulting estrangement *alienation* — from the product, the process, fellow workers and himself
Time	By season, daylight and task	By the clock and the shift — punctuality becomes a moral virtue taught by school and factory alike
Status in family	Age and seniority command	Wage-earning brings the young and women bargaining power; authority of elders declines

The balanced verdict: the Industrial Revolution multiplied wealth, goods, knowledge and eventually health and leisure beyond anything earlier ages knew — and it did so through a transition that was

brutal for the generations who lived it. Both halves are true, and sociology exists in large part to explain this double face: nearly every later topic in this course — stratification, urbanization, population policy, development — is a footnote to it.

6. Glossary — key terms of this topic

TERM	MEANING
Factory system	Production by wage workers gathered in one workplace, using powered machinery owned by an employer.
Domestic system	The pre-factory arrangement in which work was done in workers' homes with their own simple tools.
Division of labour	Breaking production into narrow specialised tasks performed by different workers.
Bourgeoisie / proletariat	The class owning the means of production / the class owning only its labour power (Marx's terms).
Alienation	Marx's term for the worker's estrangement from product, process, fellow workers and self under machine production.
Trade union	An organised association of workers formed to bargain collectively over wages and conditions.

7. Key points and summary

- The Industrial Revolution (Britain, from c. 1760) replaced hand production in homes with powered machine production in factories — beginning with cotton textiles.
- Its stages run from steam and mechanisation, through electricity and mass production, to computer automation and today's Industry 4.0.
- Economic effects: the factory system, mass production, capitalism's modern forms, division of labour, world markets — and unemployment and pollution.
- Social effects: explosive urban growth, new classes, harsh early conditions answered by factory law and unions, rapid population growth.
- Family and work were transformed together: home and workplace separated, the nuclear family rose, women's wage work began, and clock-time and alienation entered ordinary life.

8. Practice

Answer in your own words; brief model answers follow.

1. Why did the Industrial Revolution begin in textiles, and why in Britain? Give three reasons for each.
2. Build a two-column comparison of the First and Second Industrial Revolutions under: energy, emblematic invention, scale of the firm.
3. “The Industrial Revolution separated home from work.” Trace three consequences of this one change for family life.
4. Explain alienation in your own words, using a sewing operator on a high-speed line as the example.
5. A handloom weaver of 1820 and a power-loom worker of 1850 produce the same cloth. Compare their situations under: tools owned, pace of work, skill, security.
6. State one argument for calling the change a “revolution” and one for calling it an evolution.

Answer key

1. Textiles: enormous and growing demand for cloth; a sequence of bottlenecks (weavers outrunning spinners, then spinners weavers) rewarded each invention; cotton was light, standardised and suited to machines. Britain: coal and iron at hand; capital from trade and colonies supplying raw cotton and markets; labour, patents and transport (canals, ports) available.
2. Energy: steam/coal vs electricity and oil. Invention: spinning machinery and the steam engine vs steel processes, the assembly line, telegraph/telephone. Firm: the single factory of hundreds vs corporations of tens of thousands across countries.
3. Any three: earnings replace shared household production, so members depend on individual wages; children are schooled rather than worked beside parents; the nuclear household fits a mobile workforce; elders lose economic authority; childcare becomes an organised problem when both parents work away.
4. The operator repeats one seam all day at the line's pace: she never makes a whole garment (product), cannot vary her method or speed (process), competes with rather than cooperates alongside neighbours (fellows), and experiences her own skill and attention as bought time rather than expression (self). That fourfold estrangement is alienation.
5. Handloom weaver: owns loom, sets own pace, holds the whole craft skill, but income insecure and falling as factory cloth undercuts him. Power-loom worker: owns nothing, pace set by machine and supervisor, skill narrowed to tending, but wage regular — while health and hours are worse and dismissal instant.
6. Revolution: within two generations it overturned production, residence, class and family — a total change of social order. Evolution: it took the better part of a century, grew out of earlier trade, agriculture and inventions, and proceeded unevenly, region by region and trade by trade.